

Media Optimization: Backend Cheat Sheet

Baseline 2026-10-07. Images, BlurHash, video, streaming, pipelines. Quick reference only: verify against the official docs and the full pages. Ladder values from the Apple HLS Authoring Specification.

FORMATS

Format	Lossy / alpha / anim	Status, licence
JPEG	lossy / no / no	universal fallback
PNG	lossless / yes / APNG	universal; graphics
WebP	both / yes / yes	wide support; BSD libwebp
AVIF	both, HDR / yes / yes	wide support; royalty-free; slow encode
JPEG XL	both / yes / yes	uneven browser support; check first
HEIC	lossy / yes / seq.	HEVC patents; iOS input, convert on ingest

Default: AVIF + WebP + JPEG fallback in <picture>. Start quality: JPEG 75-85, WebP 75-80, AVIF 50-65 (libvips). Scales differ per encoder: judge by eye and VMAF.

VARIANTS AND STORAGE

- Widths 320 / 640 / 1280 (+1920), ~1.5x steps; never upscale.
- Crop in the backend: cover, contain, smart crop (libvips attention).
- Keys: covers/<id>/w640.avif; add a hash when content can change.
- Immutable: Cache-Control: public, max-age=31536000, immutable; Vary: Accept if negotiating.
- Apply EXIF rotation, convert to sRGB, strip EXIF/GPS, keep the original private.
- API returns width, height, blurhash, variant list.

```
vips thumbnail in.heic
out.avif[Q=60,effort=5] 1280 --size down
cwebp -q 78 -m 5 -resize 1280 0 in.png -o
out.webp
avifenc --min 20 --max 35 --speed 6 in.png
out.avif
```

BLURHASH

String: 1 size flag + 1 max AC + 4 DC + 2 per AC. Length 4 + 2 nx ny (28 for 4×3).
Size flag = (nx-1) + 9(ny-1). Alphabet: base 83 (0-9A-Za-z#\$%*+, - . : ; = ? @ [] ^ _ { } ~).
Linear light: $v = b/255; v/12.92$ if $v \leq 0.04045$ else $((v+0.055)/1.055)^{2.4}$.
Component: $C_{ij} = N/(WH) \sum \cos(\pi i x/W) \cos(\pi j y/H) L(x,y)$; $N=1$ for (0,0), else 2.
Max AC: $q = \text{clamp}(\text{floor}(166 A - 0.5), 0, 82)$; $M = (q+1)/166$.
AC channel: $q = \text{clamp}(\text{floor}(9 \text{ signPow}(c/M, 0.5) + 9.5), 0, 18)$; pack $19^2 qR + 19 qG + qB$ (2 digits).
Decode: $c = \text{signPow}((q-9)/9, 2) \cdot M \cdot \text{punch}$; $L(x,y) = \sum C_{ij} \cos \cos$.

- Encode from ~64 px; decode at 20-32 px; 4×3 default.
- TS decoder: punch | 1 truncates, then sets the low bit (2→3; only odd integers work).
- Alt: ThumbHash (alpha, aspect ratio).

FFMPEG COMMANDS

```
ffmpeg -i in.mov -c:v libx264 -preset slow -crf 22 -pix_fmt yuv420p -c:a aac -b:a 128k -movflags +faststart o.mp4
ffmpeg -i in.mov -c:v libx265 -crf 26 -tag:v hvc1 -c:a aac o.mp4
ffmpeg -i in.mov -c:v libsvtav1 -preset 6 -crf 32 -c:a libopus o.mp4
ffmpeg -i in.mov -c:v libvpx-vp9 -crf 32 -b:v 0 -c:a libopus o.webm
# capped CRF, 2 s keyframes at 24 fps
-vf scale=-2:720 -maxrate 3000k -bufsize 6000k -g 48 -keyint_min 48 -sc_threshold 0
# covers
ffmpeg -ss 5 -i in.mp4 -frames:v 1 -q:v 2 cover.jpg
-vf "thumbnail=n=300,scale=720:-2" -frames:v 1 c.jpg
-vf "fps=1/10,scale=160:-2,tile=10x10" -frames:v 1 s.jpg
# HLS fmp4
-f hls -hls_time 6 -hls_playlist_type vod -hls_segment_type fmp4
```

LADDER (APPLE HLS SPEC, KBIT/S)

H.264	kbit/s	HEVC	kbit/s
416×234	145	640×360	145
640×360	365	768×432	300
768×432	730 / 1100	960×540	600 / 900 / 1600
960×540	2000	1280×720	2400 / 3400
1280×720	3000 / 4500	1920×1080	4500 / 5800
1920×1080	6000 / 7800	2560×1440	8100
		3840×2160	11600 / 16800

- Keyframe every 2 s; segments 6 s starting on an IDR.
- Default rung: 2000 kbit/s H.264. VOD peak ≤ 200% of average.
- VP9/AV1 rungs are assumptions: start from HEVC, verify with VMAF.

STREAMING

- MP4 + faststart for short clips; HLS / DASH for the rest.
- CMAF (fmp4) serves HLS and DASH from one set of files.
- Playlists: master (EXT-X-STREAM-INF) + one media playlist per rendition.
- Private video: signed cookies on a path prefix, not per-file signed URLs. CORS for the player origin.
- DRM: Widevine, FairPlay, PlayReady via a packager; otherwise unencrypted ABR + signed cookies.
- Packagers: Shaka Packager, Bento4.

LIBRARY PICKER

Lang	Images	Video
Java	Thumbnailator (MIT); imgscalr: unmaintained, 2012 ; TwelveMonkeys (WebP read only); Scrimage; vips-ffmpeg	Jaffree (Apache-2.0); JAVE2: GPL-3.0
Python	Pillow (AVIF built in 11.3+), pillow-heif, pyvips	subprocess; MoviePy 2; ffmpeg-python: unmaintained, 2019
Node	sharp (libvips, Apache-2.0); Jimp (pure JS, slow)	spawn ffmpeg; fluent-ffmpeg: archived May 2025
.NET	ImageSharp: Split License, paid above US\$1M ; SkiaSharp (MIT); NetVips	FFmpegCore (MIT); Xabe.FFmpeg: non-commercial
Rust	image crate; blurhash crate; libvips	tokio::process + ffmpeg

PIPELINE CHECKLIST

- Never transcode in the request.
- Pre-signed PUT to a private uploads/ prefix (size and type pinned).
- Storage event or /complete call enqueues a job (ids only).
- Worker: validate magic bytes, caps on size/duration/pixels; variants, BlurHash, renditions; write outputs to fixed keys.
- Idempotent; retry with backoff; dead-letter queue; alert on queue age.
- Status uploaded → queued → processing → ready | failed; poll or webhook.
- Sandbox FFmpeg: no network, non-root, CPU/mem/time limits; patch libs.
- Lifecycle rules on uploads/; CDN for media/.